

Proactive VLSI Congestion Prediction Using a Hybrid Residual U-Net with Squeeze-and-Excitation Blocks

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Abstract - Routing congestion is a critical challenge in modern Very Large-Scale Integration (VLSI) design, as it directly impacts timing closure, power efficiency, and manufacturability. In the placement and routing (P&R) stage of VLSI design, accurate congestion prediction is essential to ensure routability, timing closure, and efficient resource utilization. Traditional Electronic Design Automation (EDA) tools rely on computationally expensive placement and routing iterations to identify congestion hotspots, which prolongs design closure cycles and increases resource consumption. To address this problem, we propose a deep learning-based framework for early congestion prediction using features derived from the CircuitNet-N14 benchmark dataset. Our approach integrates RUDY, RUDY_pin, and macro-region features with ground truth congestion maps to construct a rich feature representation of circuit layouts. A Residual U-Net enhanced with Squeeze-and-Excitation (SE) blocks is employed to learn mappings between placement features and congestion heatmaps. Experimental results on multiple benchmark designs demonstrate that our model achieves competitive accuracy, with a Structural Similarity Index (SSIM) of 0.9724 and a normalized RMSE (nRMSE) of 0.0123, while significantly reducing prediction time compared to conventional EDA flows. These results show that our framework can serve as a fast, accurate proxy for costly P&R iterations, helping the industry accelerate design closure and reduce computational overhead.

Index Terms - Electronic Design Automation (EDA), Placement and Routing (P&R), Congestion Prediction, Residual U-Net, Squeeze-Excitation (SE) Blocks, Structural Similarity Index (SSIM), Normalized Root Mean Square Error (NRMSE).

I. INTRODUCTION

The placement and routing (P&R) stage is one of the most critical steps in chip design. After logic synthesis, designers need to place the netlist onto the silicon floorplan and connect the components through global and detailed routing. These steps play a major role in determining wirelength, timing, congestion, power distribution, and ultimately whether the design can be manufactured successfully. Any bottleneck in P&R—especially routing congestion—can cause timing violations, high power consumption, or even make the design impossible to route [1]. Traditionally, Electronic Design Automation (EDA) tools handle congestion by repeatedly running placement, routing, and congestion analysis. Designers then tweak the placement or floorplan and repeat the process until congestion is acceptable. While this iterative approach

eventually works, it is slow, computationally expensive, and offers little opportunity to catch problems early. As technology nodes become smaller and more complex, this inefficiency grows, highlighting the need for faster, more predictive methods [2], [3]. The limitations of this trial-and-error paradigm highlight the need for predictive methods capable of identifying routing bottlenecks early in the flow. Recent advances in machine learning (ML) offer a promising alternative. ML models can predict congestion directly from placement features. This allows designers to identify and fix potential issues much earlier in the flow. Convolutional neural networks (CNNs) are particularly effective at capturing spatial layout patterns, and with open datasets like CircuitNet [4]. Researchers now have benchmarks to train and evaluate these models [5]. Moreover, recent works extend beyond pure congestion forecasting to explicitly optimize routability during placement [6] and apply hybrid learning strategies for improved prediction fidelity [8]. Despite these advances, challenges persist in hotspot localization, balancing numerical accuracy with visual similarity, and maintaining industrial-scale efficiency [3], [8]. To address these gaps, this study introduces a hybrid framework combining Residual UNet with Squeeze-Excitation blocks. The key contributions are:

- Industrial relevance by providing a scalable, computationally efficient predictor that reduces costly P&R iterations, enhancing real-world applicability.
- Higher predictive accuracy outperforming baseline models in both quantitative and visual metrics. We combine normalized Root Mean Squared Error (nRMSE), Structural Similarity Index (SSIM), and gradient loss to enhance both numerical accuracy and visual similarity of predicted congestion maps.

The remainder of this paper is organized as follows: Section II literature review, Section III explains methodology, Section IV presents result analysis and Section VI brings conclusion.

II. LITERATURE REVIEW

A. Related Work

Routing congestion has long been a bottleneck in physical design. Early approaches, such as Kahng et al.'s RePlAce [1], integrated routability-driven objectives into placement to reduce post-routing congestion. Similarly, Chan et al. [2] investigated ML-assisted routability optimization at sub-14nm

nodes, highlighting the growing difficulty of congestion management in advanced technologies. With the rise of ML, congestion prediction shifted from heuristic-driven to data-driven approaches. Xie et al. [8] proposed RouteNet, a CNN-based framework that effectively captured spatial dependencies in placement layouts for routability prediction. Liu et al. [6] extended this by integrating deep learning into global placement, achieving explicit routability optimization. More recently, Choi et al. [7] applied ML ordering strategies in global routing, demonstrating improvements in congestion management efficiency. Datasets have been instrumental in advancing this research direction. Chai et al. [4] introduced CircuitNet, an open-source dataset with domain-specific evaluation metrics tailored to routability and timing, while Jiang et al. [5] accelerated routability studies using heterogeneous platforms. Cho et al. [9] reviewed broader applications of ML across EDA workflows, emphasizing its transformative potential in routability analysis and beyond. Despite these advances, challenges remain. Lin et al. [3] demonstrated that hotspot prediction at sub-14nm nodes remains difficult due to increased design complexity, while Alawieh et al. [6] showed that large-scale FPGA congestion prediction still demands significant computational overhead. Moreover, existing ML-based predictors often focus on either numerical accuracy or perceptual similarity of congestion maps, rarely achieving both simultaneously [4]. Researchers further explored ML-based congestion prediction in advanced technology nodes, highlighting techniques that improve hotspot localization and reduce computational overhead by NRMSE value of 0.04 and SSIM value of 0.79 [7].

B. Research gap

Although machine learning has advanced congestion prediction in VLSI design, current approaches still face significant limitations. For instance, Liu et al. [6] and Xie et al. [8] applied CNN-based models to predict congestion, achieving high numerical accuracy but often compromising the perceptual quality of congestion maps, which limits interpretability of fine-grained hotspots. Lin et al. [4] and Chan et al. [2] highlighted that precise hotspot localization remains challenging, particularly at sub-14nm nodes, due to increased routing density and layout complexity. Furthermore, Chai et al. [4] and Jiang et al. [5] emphasized the importance of including placement aware contextual features—such as macro-block distributions, pin densities, and regional layout characteristics—but noted that many current models underutilize these features, reducing early prediction effectiveness. These limitations collectively indicate the need for a predictive framework that integrates diverse placement features, achieves both high numerical accuracy and perceptual fidelity, localizes congestion hotspots precisely, and remains computationally efficient, thereby enabling proactive congestion mitigation and more efficient design closure in advanced technology nodes.

The proposed Residual UNet + Squeeze-Excitation hybrid model achieved an NRMSE of 0.0123 and SSIM of 0.9724.

III. METHODOLOGY

Designers follow the standard VLSI chip design flow, which begins with system specification, architectural design, and logic synthesis, progressing through physical design stages such as partitioning, floor planning, and placement.

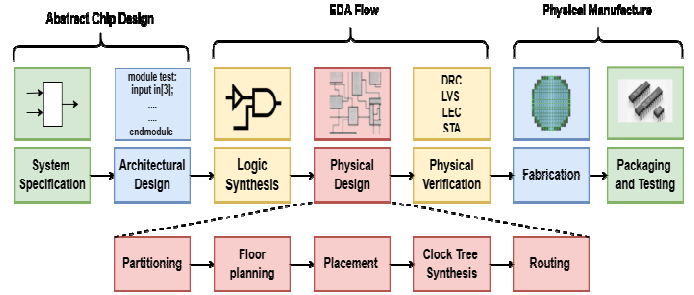


Fig. 1: Illustration of VLSI chip design flow [2]

The Fig. 1 illustrates the VLSI chip design flow, from system specification to packaging and testing. After the placement stage, where components are arranged on the chip layout, congestion prediction is performed using our deep learning model. This model, based on a ResNet architecture enhanced with Squeeze-and-Excitation (SE) blocks, effectively identifies potential congestion hotspots in the design, ensuring optimal routing and overall chip performance.

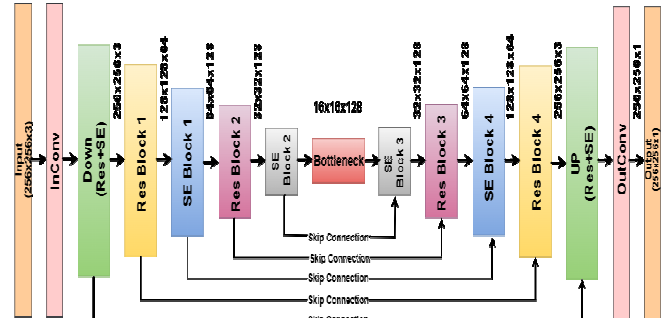


Fig. 2: Overall methodology of Residual U-Net and SE block hybrid model

The proposed model in Fig. 2 employs a Residual U-Net backbone enhanced with Squeeze-and-Excitation (SE) blocks for channel attention. The encoder progressively extracts hierarchical placement features through convolution, down sampling, and residual connections, while the bottleneck captures global spatial dependencies. The decoder, integrated with skip connections and SE refinement, reconstructs accurate congestion maps with improved hotspot localization and structural similarity.

A. Dataset

CircuitNet-N14 [14] is an open-source dataset based on the 14nm FinFET process node. The dataset provides rich features at the cell, net, and region levels, including placement density, wire length, fan out, macro placement, and power grid information. Fig. 3 shows a visualization of the dataset.

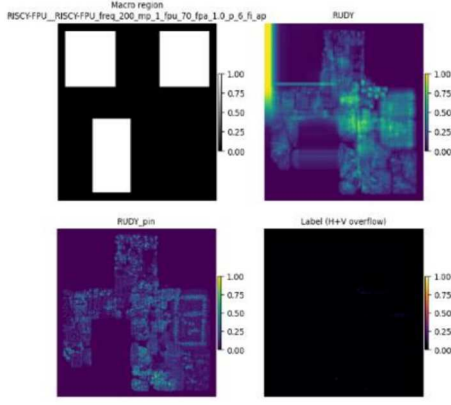


Fig. 3: Visualization of the dataset

The dataset also offers grid-based 2D maps for congestion, IR drop, and delay, enabling CNN- and GNN-based modelling. With over 500 designs and high-resolution data (64×64 to 256×256 grids), CircuitNet-N14 serves as a benchmark for tasks like congestion prediction, routability analysis, and power integrity evaluation in advanced VLSI.

Table 1 shows the statistics like number of cells, nets, count of macros, and amount of samples of the designs that are used for training and testing the model. All the samples have been merged into a single dataset and trained the model.

Table 1: Statistics of designs inCircuitnet-14 [14]

Design	Cells	Nets	Macros	Samples
RISCY	46,184	47,233	3	3,456
RISCY-FPU	65,464	66,903	3	3,456
Vortex-small	113,961	124,058	43	96
NVDLA-small	270,072	285,465	108	89

B. Data Preparation

The proposed methodology formulates congestion prediction as a supervised learning problem, where the input feature tensor is defined as,

$$X = [X^{(m)}, X^{(r)}, X^{(p)}] \in [0,1]^{H \times W \times 3} \quad (1)$$

Where, $X^{(m)}$ =macro mask , $X^{(r)} = RUDY_{pin}$. Targets are horizontal/vertical overflow maps $Y^{(h)}, Y^{(v)} \in R_+^{H \times W}$. Resize operator R_{256} maps inputs to 256×256 (nearest for $X^{(m)}$, area for others). The dataset was split into 85% training, 10% validation, and 5% testing subsets, ensuring coverage across multiple architectures such as CPUs, GPUs, and accelerators. The ground truth congestion map is obtained by summing horizontal and vertical overflow maps followed by MinMax normalization,

$$\tilde{X}^{(c)} = \begin{cases} 1\{R_{256}(X^{(m)}) > 0.5\}, & c = m, \\ \frac{R_{256}(X^{(c)}) - \min}{\max - \min}, & c \in \{r, p\}, \end{cases} \quad (2)$$

$$\tilde{Y} = \text{MinMax} \left(R_{256}(Y^{(h)}) + R_{256}(Y^{(v)}) \right) \in [0,1]^{H \times W} \quad (3)$$

Where $\tilde{X}^{(c)}$ = Normalization function.

C. Squeeze and Excitation (SE) Block

The SE block adaptively recalibrates channel-wise feature responses by explicitly modelling interdependencies between channels. Let the input feature map be $X \in \mathbb{R}^{H \times W \times C}$ the operations are as follows:

$$Z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j) \quad (4)$$

$$s = \sigma(W_2 \cdot \delta(W_1 \cdot Z)) \quad (5)$$

$$\hat{X}_c = s_c \cdot X_c \quad (6)$$

Where Z_c = Squeeze , s = Excitation , $\hat{X}_c$ =Recalibration, C =channel, H =height , W =width, W_1 and W_2 = trainable weight matrices of two Fully Connected (FC) layers inside the excitation step, δ = ReLU function and σ = sigmoid.

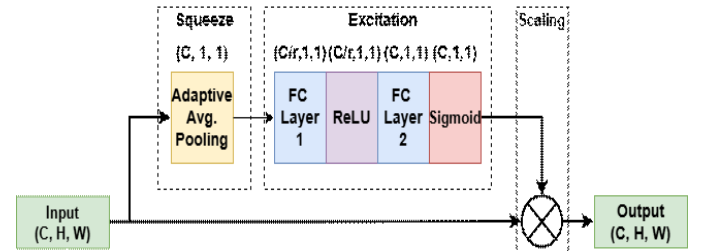


Fig. 4: Illustration of Squeeze and Excitation (SE) block

Thus, the SE block learns channel attention by applying the scaling vector s to the input feature maps, enhancing informative channels and suppressing less useful ones. Fig. 4 illustrates the SE block.

D. Network f_θ : Residual U-Net and SE block

The predictive model, f_θ is constructed using a Residual U-Net augmented with Squeeze-and-Excitation (SE) blocks, where each feature transformation is defined as $\text{ResSE}(Z)$, with $F(\cdot)$ representing a Conv-BN-ReLU stack, GAP global average pooling, δ the ReLU function, and σ the sigmoid. Down sampling is performed using strided convolutions, while up sampling employs bilinear interpolation followed by convolution, with skip connections refined through ResSE fusion. Model training minimizes a composite objective balancing pixel-wise accuracy, structural similarity, and boundary preservation.

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{NRMSE}} + \beta \cdot (1 - \text{SSIM}) + \gamma \cdot \mathcal{L}_{\text{EDGE}}$$

Where $\mathcal{L}$ = composite loss , $\mathcal{L}_{EDGE}$ = edge loss , α, β, γ = weights. The algorithm for this process is given below:

Algorithm 1: Prediction Framework using Residual U-Net and SE block

Input: Placement Features X
Output: Congestion heatmap $\hat{Y}$

1. Download dataset CircuitNet-14
2. Extract placement feature:
 $X \rightarrow F_0 = \{R_{256}(X^{(m)}), R_{256}(X^{(r)}), R_{256}(X^{(p)})\}$
3. Generate training set
4. Define Model (Residual UNet + SE blocks)
 5. Define SEBlock: $SE \leftarrow \sigma(W_2 \delta(W_1 \cdot GAP(Z))) \odot Z$
 6. Define ResBlock : $ResSE(Z) \leftarrow SE(F(Z)) + Z$
 7. Define Down: $Z_{l+1} \leftarrow ResSE_l(Conv_{3 \times 3, s=2}(Z_l))$
 8. Define Up: $\tilde{Z}_l \leftarrow Conv_{2 \times 2}(Upsample(Z_{l+1}))$
 9. Skip Connection with SE:
 $Z'_l = ResSE_l(\text{concat}(\tilde{Z}_l, Z_l))$
 10. Final UNetSE output: $\hat{Y} \leftarrow f_{\theta}(X)$
 11. Define Composite Loss, $\mathcal{L}$
 12. Define evaluation metrics (SSIM , NRMSE)
13. **for** epoch in range **do**
 14. model.train()
 15. **for** each batch $(X, Y) \in \text{train_loader}$ **do**
 16. Forward Pass: $\hat{Y} \leftarrow f_{\theta}(X)$
 17. Compute $\mathcal{L}$
 18. Backward + optimizer step (AMP)
 19. Evaluate on validation set:
 $\mathcal{L}_{val}, SSIM_{val}, NRMSE_{val}$
 20. **end for**
 21. Update scheduler
 22. Save Checkpoint $\leftarrow \mathcal{L}_{val}$ if improved
 23. **end for**
24. Repeat until convergence.

The Fig. 5 illustrates the full pathway a data will flow in the model. The data after preprocessing stage enters into feature extraction procedure. Later on, Residual U-Net + SE block handles the data for loss computation and finally predicted congestion map is obtained as output.

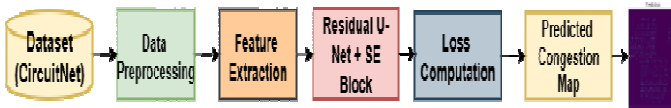


Fig. 5: Workflow of the model

E. Model Hyper Parameters

The ResUNet-SE model was trained using the Adam optimizer and a batch size of 8 for 60 epochs. The loss function combined MSE (weight 1.0) for pixel-wise accuracy and SSIM (weight 0.5) for structural consistency. Total number of Trainable parameters are 12187045.

The optimum hyper parameter values for the proposed model are shown in table 2

Table 2: Hyper parameters utilized to train the model.

Hyperparameters	Optimal Value
Optimizer	Adam W
Dropout	Not Used
Learning Rate	Adaptive
Batch Size	8
Epoch	60
MSE weight	1.0
SSIM weight	0.5
Base Channel	48
Input Channel	3
Output Channel	1
SE Reduction Ratio	8

The network had 3 input channels, 1 output channel, and a base channel size of 48, with an SE reduction ratio of 8 for channel attention. These hyper parameters were selected to ensure stable training and accurate congestion map prediction.

IV. RESULT ANALYSIS

The evaluation is based on multiple metrics, including SSIM (Structural Similarity Index), NRMSE (Normalized Root Mean Squared Error), and visual error analysis. The train composite loss in Fig. 6 shows a steady decrease, particularly after the first few epochs, indicating that the model effectively learns the congestion patterns.

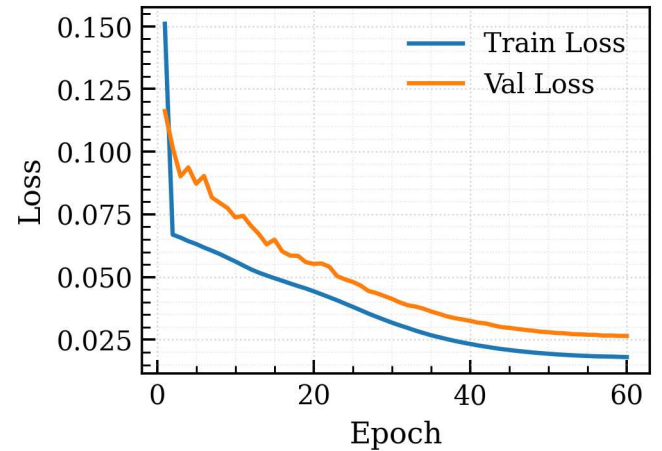


Fig. 6: Train composite loss curve

The SSIM curves in Fig. 7 starts from a lower value and gradually increases to 0.9724 reflecting the model's improved capability to capture fine-grained structural information over time.

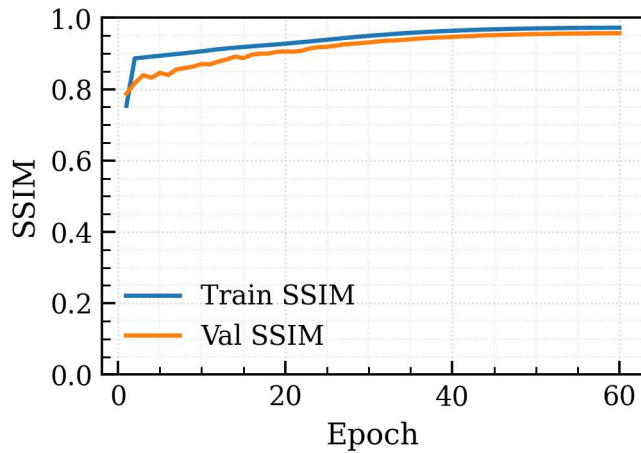


Fig. 7: Train SSIM curve

The NRMSE curve in Fig. 8 steadily improves (decreases), indicating that the model's prediction accuracy for congestion is improving over the epochs. The best train NRMSE of was 0.0123.

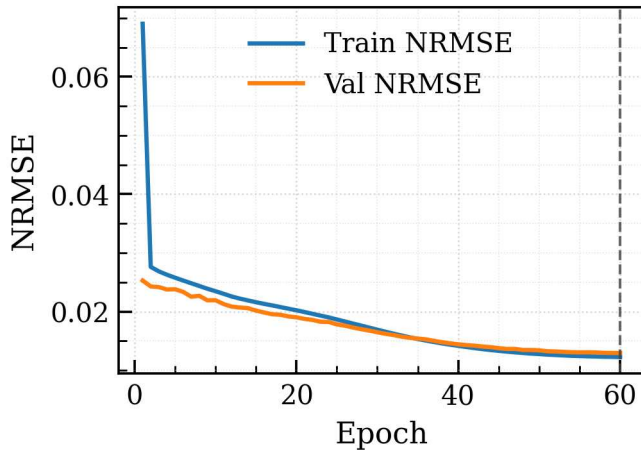


Fig. 8: Validation NRMSE curve

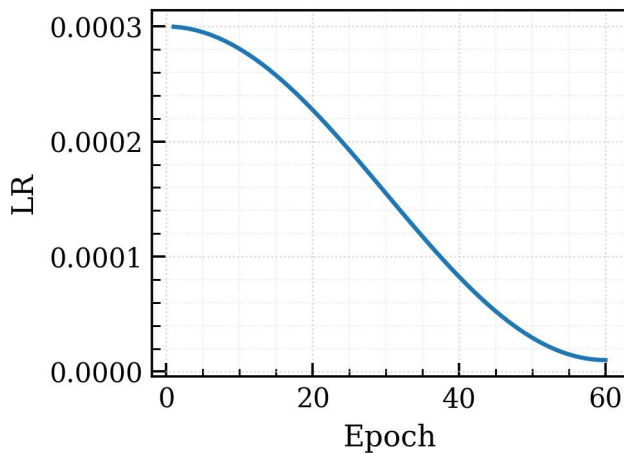


Fig. 9: Learning Rate curve

The adaptive learning rate (LR) in Fig. 9 follows a cosine annealing schedule, starting at 0.0003 and gradually decreasing

throughout training. This strategy helps in stabilizing the learning process as the model converges.

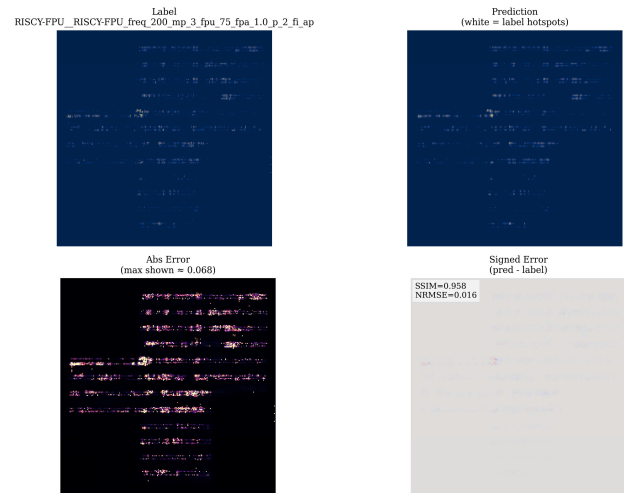


Fig. 10: Performance of Proposed Model

The validation and test predictions in Fig. 10 show the model's ability to accurately predict congestion in circuit designs. The absolute error maps reveal minimal discrepancies between the predicted and actual congestion, with dark areas indicating low error. Signed error maps highlight areas of over- and under-prediction, while the SSIM and NRMSE metrics show excellent structural similarity and low error, confirming the model's effectiveness. Table 3 shows the performance indicators associated with the models. ↑ indicates “higher is better” and ↓ indicates “lower is better”.

Table 3: Performance indicators associated with the models

Model	SSIM ↑	NRMSE ↓
sFCN+ CNN	0.5890	0.0525
Tiny U-Net	0.6984	0.0509
SegNet	0.7452	0.0467
SegNet + SE Block	0.7553	0.0499
Residual U-Net + SE Block (Proposed)	0.9695	0.0123

The proposed model combining Residual U-Net and Squeeze and Excitation (SE) block shows superior SSIM and NRMSE values compared to existing models.

Table 4 shows the comparison results with respect to SSIM and NRMSE.

Table 4: Comparison results with respect to SSIM and NRMSE

Author	Model	SSIM	NRMSE
S. Zheng et al. [13]	Lay-Net	0.7170	0.0720
Q. Cai, Q. Cui, Z. Cheng [11]	CongestionNet (sFCN with MixConv)	0.7375	0.0514
H. Yang et al.[14]	RouteNet	0.7805	0.0493
H. Li et al.[12]	Inception-Boosted U-Net	0.7940	0.0467
X. Jiang et al.[5]	CircuitNet Model	0.8000	0.0400
PROPOSED	Residual UNet + Squeeze-Excitation	0.9567	0.0123

In conclusion, the proposed Residual U-Net with SE blocks offers strong performance in predicting congestion in circuit design, achieving high SSIM scores and low NRMSE values by outperforming previous models particularly maintaining accurate predictions.

V. CONCLUSION AND FUTURE WORK

In this work, we have developed a deep learning-based framework for early congestion prediction in the placement and routing (P&R) stage of VLSI design. By leveraging features from the CircuitNet-N14 dataset and employing a Residual UNet enhanced with Squeeze-and-Excitation (SE) blocks, we achieved significant reductions in prediction time compared to traditional EDA tools. Our model demonstrated competitive performance, with an SSIM of 0.9724 and an nRMSE of 0.0123, highlighting its accuracy and visual similarity. This framework offers a scalable and computationally efficient alternative to costly iterative P&R processes, enabling faster design closure and optimized resource utilization. In future, we aim to integrate the proposed framework into existing Electronic Design Automation (EDA) toolchains, enabling seamless industrial deployment and real-time congestion prediction during the placement and routing stages. This integration will facilitate efficient design closure, minimize computational overhead, and streamline the overall VLSI design process.

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